

Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: B.E., V & ECE B

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.G.Sivakumar

Innovative Practice Description

- **Unit / Topic:** Unit IV/Transmission Control Protocol
- **Course Outcome:** Students will be able to explain the flow of information from one node to another node in the network.
- **Topic Learning Outcome:** Explain the concept of Transmission Control Protocol and its services.
- **Activity Chosen:** Jigsaw
- **Justification:**

The activity 'JIGSAW' is used to recall the concepts learnt and enhance the learning through this collaboration. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Transmission Control Protocol is an important topic and having subdivisions, so the activity 'JIGSAW' is used to recall the concepts learnt and enhance the learning through this collaboration.

Time Allotted for the Activity: 1 Hour 30 minutes

• **Details of the Implementation:**

The lessons are divided into subcategories. Students are divided into groups of 6 or 7 Students. The Class Strength is divided into 7 Groups with 6 members. The 7 groups are named as 1,2,3,4,5,6,7. Each group allotted with one topic, student learns about his or her topic, preparation materials are posted in CANVAS and presents it to group. Next, Expert group is formed consist of members from each group and discuss their topic with its member 15 Minutes, which has been already discussed with home group. Next, students gather into groups divided by topic. Expert group member presents topics to home group members. It is a cooperative learning method that brings about both individual accountability and achievement of the team goals. Each of these group is given a different topic and allowed to learn about it.

CO – PO / PSO mapping:

CO	PO1	PO2	PO4	PO8	PO9	PO10	PSO1
CO4	2	2	1	1	1	1	2

PO / PSO mapped:

Innovative practice	PO8	PO9	PSO1
	1	1	2
Justification for correlation	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at level 2	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	The student will be able to analyze and understand services and state transition of TCP, So it is mapped at level 2.

Images / Screenshot of the practice:



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

Students feel that it was interesting. They learn how to communicate with team members and work together.

❖ **Benefit of the practice:**

Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 80% of students answered correct

❖ ***Challenges faced in implementation:***

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ <https://www.readingrockets.org/strategies/jigsaw>
- ❖ <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>

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- **Unit / Topic:** V/ Electronic Mail
- **Course Outcome:** Students able to classify different kind of transport layer protocol
- **Topic Learning Outcome:** Explain the concept of mail transfer protocol: SMTP, POP, IMAP and MIME
- **Activity Chosen:** Flipped Class Room
- **Justification:** It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**

Specific topic was given to the students learn on their own. Resources like reference book, PPT and videos were already post in canvas. This flipped class is used for the topic apart from the learning materials given by the faculty, students have chance to browse and find more information related to Electronic mail. All students are well known about mail and easy to understand. So this topic is selected for flipped class room activity. Students are separate into groups where students are given a task to perform. Get the class back together to share the individual group's work with everyone.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12	PSO1
CO5	2	3	2	1	1	1	1	1	1	2

PO / PSO mapped:

Innovative practice	PO8	PO9	PSO1
	1	1	2
Justification for correlation	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at level 2	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	Students will be able to adapt with new development electronics mail protocols, so it is mapped with level 2.

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

Students feel that they have improved self-learning. They learn how to communicate with team members and work together.

❖ **Benefit of the practice:**

Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

❖ **Challenges faced in implementation:**

The main challenge faced is that few students not exposed to flipped class room. Students struggle with self-discipline and may turn up to class without having absorbed the lesson. To make all students to participate I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.

- ❖ Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from <https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html>.



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Department of Electronics and Communication Engineering

Academic Year: 2021-2022 (Odd Semester)

Innovative Practices Description for Unit-I

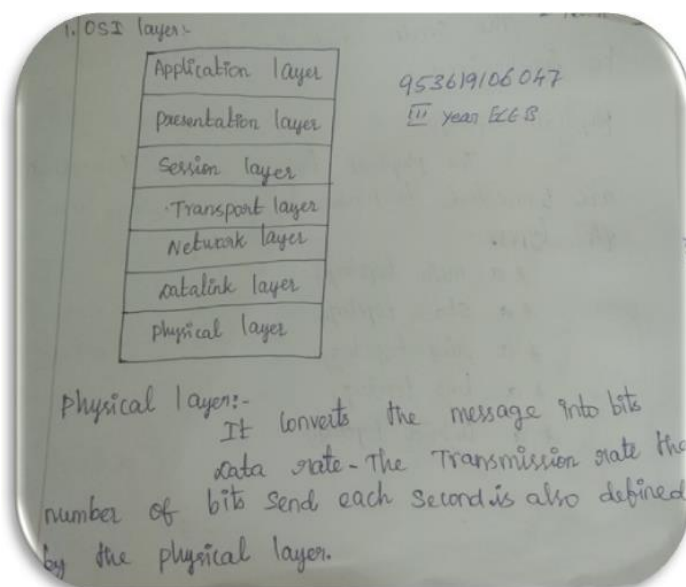
Degree, Semester & Branch: V Semester B.E. ECE B

Course Code & Title: EC8551 Communication Networks

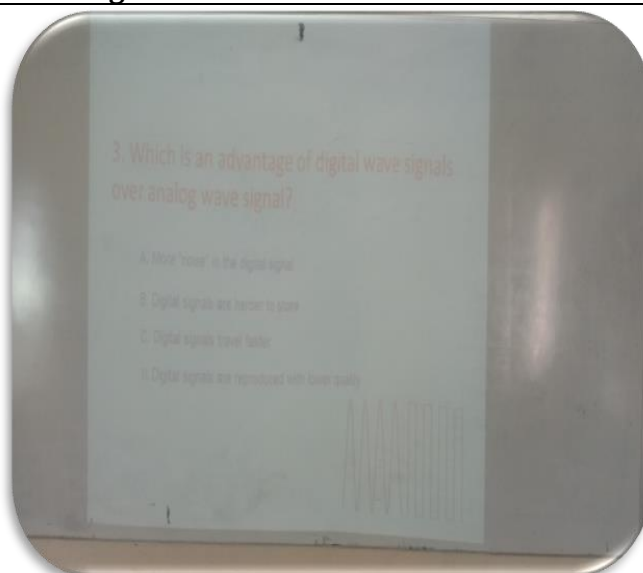
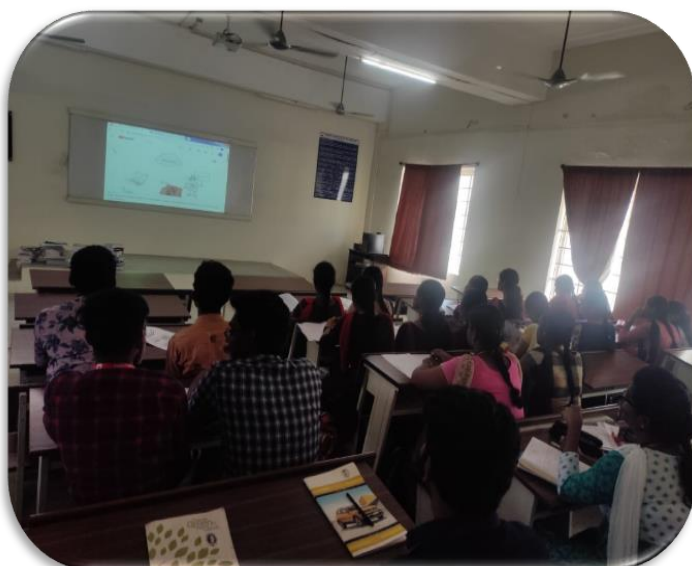
Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic	Name of the Innovative Practice	Date & Duration
OSI Model	Minute Paper	(10 Minutes)
Overview of Data and Signals	Visible Quiz	(5 Minutes)
Error Detection and Correction	Zero Minute Speech	(10 Minutes)

OSI Model



Overview of Data and Signals



Error Detection and Correction





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Innovative Practices Description for Unit-II

Degree, Semester& Branch: V Semester B.E. ECE B

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic	Name of the Innovative Practice	Date & Duration
Wireless LAN – Available Protocol	Class Poll	(5 Minutes)
Zigbee	Think Pair share	(15 Minutes)
Network Layer protocols – IP, ICMP	Story board	(10 Minutes)

Wireless LAN – Available Protocol - Class Poll



Zigbee - Think Pair share





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Innovative Practices Description for Unit-III

Degree, Semester & Branch: V Semester B.E. ECE B

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic	Name of the Innovative Practice	Date & Duration
Routing	Informal Group	(10 Minutes)
Overview of Intradomain protocols	Round Table	(5 Minutes)
Overview of IPV6 Addressing	Concept Test	(10 Minutes)

Informal Group



Round Table

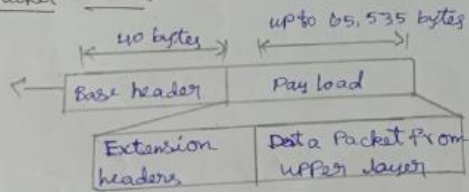


Concept Test

Internet Protocol Version 6 (IPv6) :-

- * An IPv6 address is 16 bytes (octets); it is 128 bits long.
- * IPv6 specifies hexadecimal colon notation.
- * In this notation, 128 bits is divided into eight sections, each 2 bytes in length.

Packet Format:



Three transition strategies :-



Rg.No.:

953619106074



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Innovative Practices Description for Unit-IV

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Course Code & Title: EC8551 Communication Networks

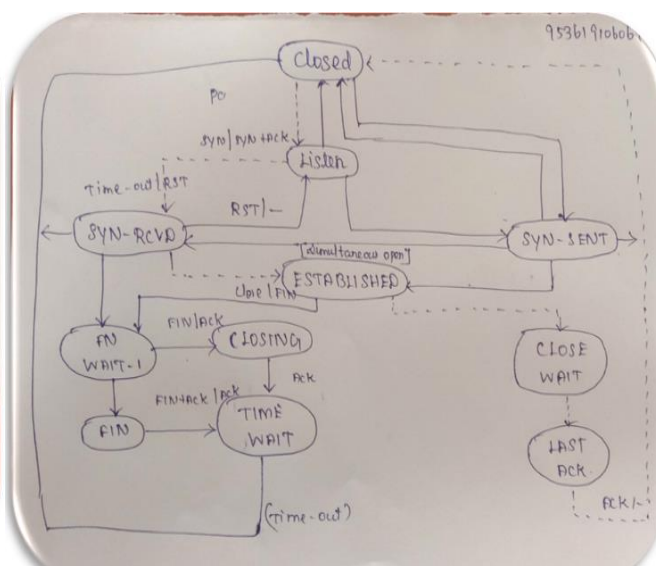
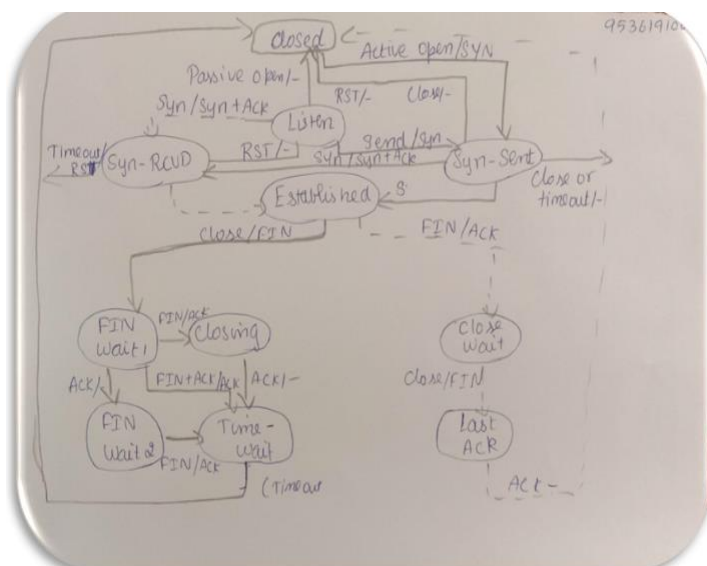
Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic	Name of the Innovative Practice	Date & Duration
Transmission Control Protocols(TCP)	Informal Group	(10 Minutes)
TCP Connection –State Transition Diagram	Mindmap	(5 Minutes)
Application requirements	Library & open book Test	(10 Minutes)

Transmission Control Protocols(TCP) - Informal Group



TCP Connection –State Transition Diagram - Mindmap



Application requirements - Library & Open book Test





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Innovative Practices Description for Unit-V

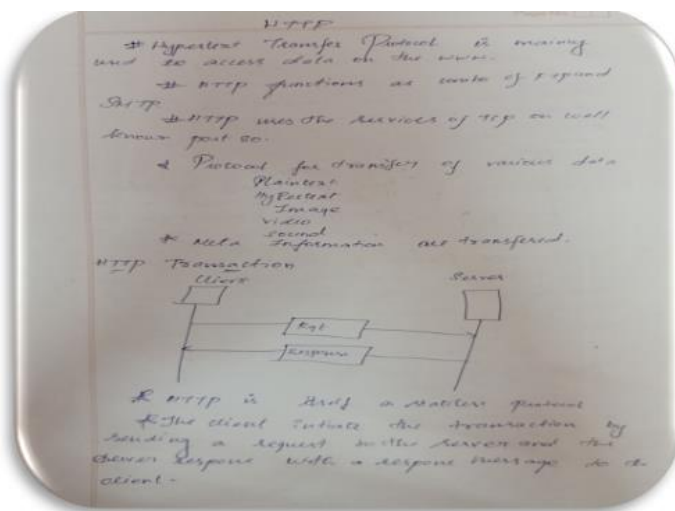
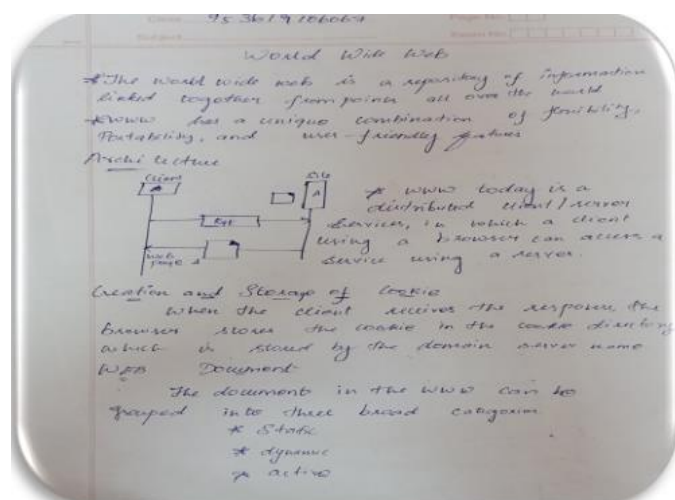
Degree, Semester & Branch: V Semester B.E. ECE B

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic	Name of the Innovative Practice	Date & Duration
WWW, HTTP	Exit Slip	(10 Minutes)
DNS	Finger Signal	(5 Minutes)
Need for Cryptography and Network Security	In-Class Team	(10 Minutes)

WWW, HTTP - Exit Slip



DNS - Finger Signal



Need for Cryptography and Network Security - In-Class Team

